Week 4 Day 5 — Executive Report

Finding high-income customers | Adult Census Income dataset | Capstone | Aqsa Saqib

1. Business goal

A marketing team wants to offer a premium product to people who earn more than $50K a year. Every contact costs money, so the goal is to reach as many real high earners as possible while wasting few contacts. I measure success with the **F1 score**, which balances wasted contacts (precision) and missed customers (recall).

2. Data used

- UCI Adult dataset (1994 US census): 32,561 people with age, education, job, marital status, working hours, capital gain and more.
- Only 24% earn more than $50K, so the classes are imbalanced.
- 80% (26,048 people) for training and 20% (6,513 people) kept as an untouched hold-out test set.
- 8 new features were added, for example *is_married*, *log of capital gain* and *education × hours*.

3. Final model & deployment artifact

- **Model:** tuned Gradient Boosting (learning_rate 0.03, 300 trees, l2_regularization 10).
- **Artifact:** one pipeline (new features → preprocessing → model) saved as *Saved_Pipelines/final_model.joblib*, together with the threshold 0.40.
- **Inference script:** *inference.py* checks the input and returns the chance, the prediction and the top 3 reasons for each person. 5 tests pass (normal input, missing column, unseen category, missing value, CSV file).

4. Key hold-out metrics

Model (hold-out test set)	Precision	Recall	F1	ROC AUC
Day 1 simple rule (Bachelors or higher)	0.498	0.511	0.505	0.674
Logistic Regression	0.745	0.622	0.678	0.909
Random Forest	0.764	0.614	0.681	0.913
Stacking (LR + RF + GB)	0.782	0.651	0.711	0.922
Gradient Boosting, threshold 0.5	0.775	0.658	0.712	0.923
Final: Gradient Boosting, threshold 0.40	**0.710**	**0.734**	**0.722**	**0.923**

- Stacking scores the same as Gradient Boosting but is slower, so it was not chosen.
- Class weight, undersampling and SMOTE did not improve the model's ranking (PR AUC). SMOTE was the worst. The imbalance is handled with the threshold instead.

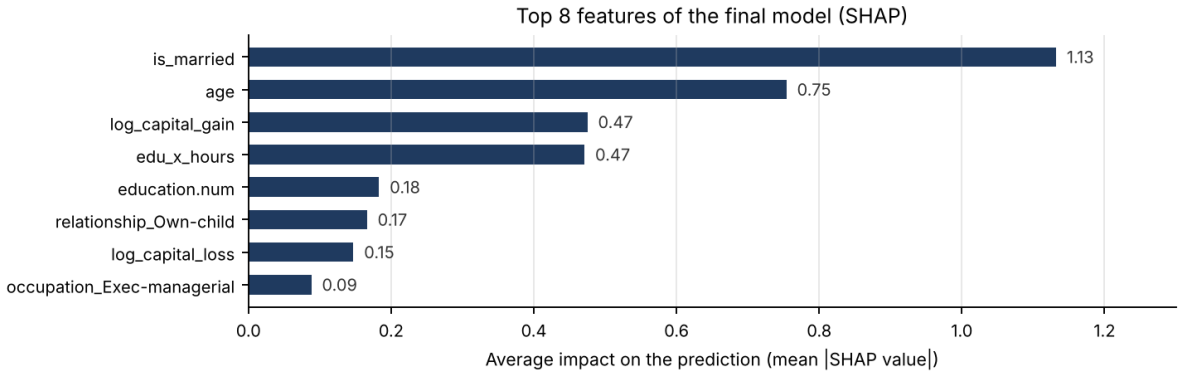
5. Chosen threshold: 0.40

The threshold was chosen on the training data with cross-validation (not on the test set), where it gave the best F1. On the test set it means, for every **1,000 people** checked:

People contacted	Real high earners reached	Wasted contacts	High earners missed
249	177	72	64

About **7 in 10** contacts are real high earners, and the model finds about **3 in 4** of all high earners. A higher threshold (0.5) wastes fewer contacts but misses more customers.

6. What drives the predictions



- **Marital status** is the strongest signal, then **age**, **capital gain** and **education with working hours**. Feature importance, permutation importance and SHAP all agree.
- Three engineered features (is_married, log_capital_gain, edu_x_hours) are in the SHAP top 4.
- **Example mistake:** a 57-year-old married manager got a chance of 0.52 but earns <=50K. Marriage, age and job pushed him up, but he has only a high-school education.

7. Fairness observations

Group (test set)	People	F1	Group (test set)	People	F1
Female	2,153	0.713	White	5,627	0.728
Male	4,360	0.723	Black	579	0.671
			Asian-Pac-Islander	196	0.653
			Amer-Indian-Eskimo	67	0.588

- **Sex:** almost no gap (F1 0.713 vs 0.723), and the share predicted >50K matches the real share for both.
- **Race:** F1 is lower for Black, Asian-Pac-Islander and Amer-Indian-Eskimo people. The smallest groups are very small in the test set, so these numbers are uncertain.
- **Mitigations:** collect more data for small groups, test the model without sex and race, check group scores every month, and do not use the model for credit or hiring decisions without a fairness review.

8. Recommended next steps

- **A/B test (4 weeks):** split the next campaign audience randomly 50/50. Group A is chosen the current way, group B by the model (threshold 0.40). Compare the conversion rate per contact and the cost per new customer.
- **Success rule:** launch the model if group B has a clearly better conversion rate (for example 10% better, statistically significant) and no new fairness gaps.
- **Monitoring:** watch input problems, data drift, the share predicted >50K (about 25%), F1 on new labelled data (alert below 0.68) and group scores. Details are in the one-page monitoring checklist.
- **Retraining:** every 6 months, or earlier if an alert lasts 2 weeks. The data is from 1994, so the model must be retrained on recent data before real use.